Extra Bite #17: Filters and Matching Networks

Pi, Pi-L, T-networks, Butterworth vs Chebyshev vs elliptical, helical and cavity and crystal filters, shape factor.

Group E7C · 11 pool questions · Do after: EC-A · 5-10 minute presentation · Practice exams at eham.net/exams

Matching Networks: Pi, T, And Pi-L

These are the three classic L-C topologies used to transform impedances and (as a side effect) filter harmonics. They are named for the shape the components make on the schematic.

Network	Topology	Q
Pi	**Capacitor from input to ground, another capacitor from output to ground, an inductor between input and output** (E7C01). The result is a low-pass network: it matches impedance at the design frequency and rolls off harmonics.	E7C01
Pi-L	**A Pi-network with an additional output series inductor** (E7C07). The extra L gives **greater harmonic suppression** than a plain Pi (E7C03).	E7C03, E7C07
T (high-pass)	**Series capacitors and a shunt inductor → high-pass** frequency response (E7C02). Same T shape, but with caps in series.	E7C02

How does an impedance-matching circuit transform a complex impedance to a resistive one? It cancels the reactive part of the impedance and changes the resistive part to the desired value (E7C04). Two jobs: tune out the reactance and transform the resistance.

Filter Families By Response Shape

These are the canonical analog-filter families. Each makes a different trade between flatness, sharpness, and ringing.

Family	Characteristic	Q
Butterworth	Maximally flat passband, gradual rolloff. (Mentioned as a trap answer, not directly tested.)	—
Chebyshev	**Ripple in the passband and a sharp cutoff** (E7C05). The price of sharper transition is wiggle inside the passband.	E7C05
Elliptical (Cauer)	**Extremely sharp cutoff with one or more notches in the stop band** (E7C06). Sharpest of the bunch; the notches push specific frequencies to deep nulls.	E7C06

Filters By Construction

The pool also tests which *physical* filter type belongs in which place.

Filter	Use	Q
Helical filter	**Most frequently used as a bandpass or notch filter in VHF and UHF transceivers** (E7C08). A coil inside a shielded cavity, tuned to the desired band.	E7C08
Crystal lattice filter	**A filter for low-level signals made using quartz crystals** (E7C09). Used in applications requiring stable, narrow bandpass filtering of low-level signals.	E7C09
Cavity filter	**Used in a 2-meter repeater duplexer** (E7C10). High-Q resonant cavities separate the close-spaced repeater TX and RX frequencies on a shared antenna.	E7C10

Shape Factor

What measures a filter's ability to reject signals in adjacent channels? Shape factor (E7C11). Defined as the ratio of the bandwidth at one stop-band attenuation level (often −60 dB) to the bandwidth at the passband edge (often −6 dB). Closer to 1 = closer to "brick wall" rolloff.

Mental Cheat Sheet

- Pi network = shunt C, series L, shunt C → low-pass matching.
- Pi-L = Pi + extra series L → better harmonic suppression.
- T with series Cs + shunt L → high-pass.
- Matching = cancel X, scale R to target.
- Chebyshev: ripple in passband, sharp cutoff.
- Elliptical: sharpest cutoff, notches in stopband.
- Helical filter → VHF/UHF bandpass/notch.
- Crystal lattice → low-level IF filter (HF SSB receivers).
- Cavity → 2-meter repeater duplexer.
- Shape factor measures adjacent-channel rejection.

Common Traps

- Pi network has **2 caps + 1 inductor**; T-low-pass has 2 inductors + 1 cap. Mixing them up is the classic mistake.
- Pi-L's extra component is an **inductor**, in series at the output.
- The Pi-L's purpose is **harmonic suppression**, not efficiency or fewer parts.
- Chebyshev has passband ripple; Butterworth is the flat one.
- Elliptical has **stop-band notches**, not stop-band ripple.
- E7C09 trap: answer C (high-Q lattice quartz crystals) sounds plausible, but the canonical defining characteristic is **low-level signals**, not Q.

- Duplexer = **cavity** filter, not crystal or DSP.
- Adjacent-channel rejection is measured by **shape factor**, not passband ripple or phase response.

Bite 17 Question Bank (E7C)

E7C01. How are the capacitors and inductors of a low-pass filter Pi-network arranged between the network's input and output?
A. Two inductors are in series between the input and output, and a capacitor is connected between the two inductors and ground.
B. Two capacitors are in series between the input and output, and an inductor is connected between the two capacitors and ground.
C. An inductor is connected between the input and ground, another inductor is connected between the output and ground, and a capacitor is connected between the input and output.
D. A capacitor is connected between the input and ground, another capacitor is connected between the output and ground, and an inductor is connected between the input and output.

E7C02. What is the frequency response of a T-network with series capacitors and a shunt inductor?
A. Low-pass.
B. High-pass.
C. Band-pass.
D. Notch.

E7C03. What is the purpose of adding an inductor to a Pi-network to create a Pi-L-network?
A. Greater harmonic suppression.
B. Higher efficiency.
C. To eliminate one capacitor.
D. Greater transformation range.

E7C04. How does an impedance-matching circuit transform a complex impedance to a resistive impedance?
A. It introduces negative resistance to cancel the resistive part of impedance.
B. It introduces transconductance to cancel the reactive part of impedance.
C. It cancels the reactive part of the impedance and changes the resistive part to the desired value.
D. Reactive currents are dissipated in matched resistances.

E7C05. Which filter type has ripple in the passband and a sharp cutoff?
A. A Butterworth filter.
B. An active LC filter.
C. A passive op-amp filter.
D. A Chebyshev filter.

E7C06. What are the characteristics of an elliptical filter?
A. Gradual passband rolloff with minimal stop-band ripple.
B. Extremely flat response over its pass band with gradually rounded stop-band corners.
C. Extremely sharp cutoff with one or more notches in the stop band.
D. Gradual passband rolloff with extreme stop-band ripple.

E7C07. Which describes a Pi-L network?
A. A Phase Inverter Load network.
B. A Pi-network with an additional output series inductor.
C. A network with only three discrete parts.
D. A matching network in which all components are isolated from ground.

E7C08. Which of the following is most frequently used as a band-pass or notch filter in VHF and UHF transceivers?
A. A Sallen-Key filter.
B. A helical filter.
C. A swinging choke filter.
D. A finite impulse response filter.

E7C09. What is a crystal lattice filter?
A. A power supply filter made with interlaced quartz crystals.
B. An audio filter made with four quartz crystals that resonate at 1 kHz intervals.
C. A filter using lattice-shaped quartz crystals for high-Q performance.
D. A filter for low-level signals made using quartz crystals.

E7C10. Which of the following filters is used in a 2-meter band repeater duplexer?
A. A crystal filter.
B. A cavity filter.
C. A DSP filter.
D. An L-C filter.

E7C11. Which of the following measures a filter's ability to reject signals in adjacent channels?
A. Passband ripple.
B. Phase response.
C. Shape factor.
D. Noise factor.

Answer Key

E7C01: D	E7C02: B	E7C03: A	E7C04: C
E7C05: D	E7C06: C	E7C07: B	E7C08: B
E7C09: D	E7C10: B	E7C11: C	